

Pelican Spray

PRODUCT BROCHURE

Technical Specifications
Features
Performance



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1. TECHNICAL SPECIFICATIONS

PROPULSION SYSTEM	
Motors	Four (4) electric motors (from s/n 8)
Power	100 kW combined
Battery	18 kWh lithium ion
Battery Charge Time	60-180 minutes
Propellers	Four (4) static propellers (from s/n 8) • Two (2) in front of wing • Two (2) behind the wing
DIMENSIONS	
Wing Span	11.5m / 38 ft
Wing Area	12.1 m ² / 130 ft ²
Length	6.0 m / 20 ft
Height	2.1 m / 7 ft
Swath Width	18 m / 59 ft @ CV approx. 15%
WEIGHTS	
Operating Empty Weight (OEM)	281 kg / 620 lbs (with Batteries) 188 kg / 415 lbs (without Batteries)
Maximum Takeoff Weight (MTOW)	599 kg / 1,320 lbs
Maximum Landing Weight (MLW)	599 kg / 1,320 lbs
CAPACITY	
Maximum Payload	317 kg / 700 lbs
Liquid Volume	318 l / 84 gal
Refill Turnaround Time	5 minutes, incl. fill, battery swap, taxi time, etc.

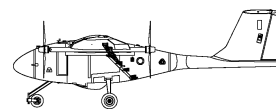
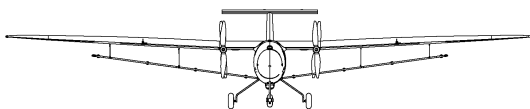
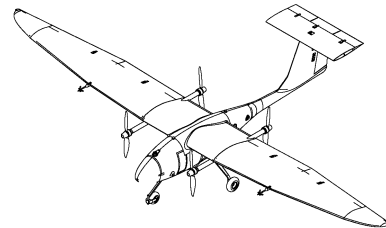
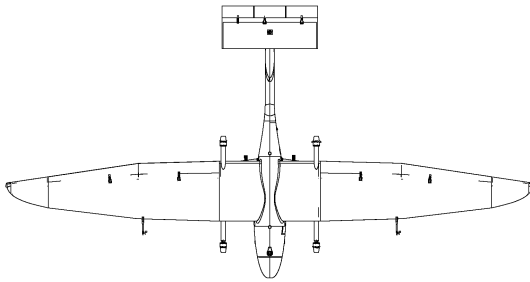
2. PERFORMANCE

SPEEDS	
Cruise Speed	60 - 70 knots / 111 - 130 km/h / 30 - 36 m/s
Stall with flaps	35 knots / 65 km/h / 18 m/s
Application Speed	60 - 70 knots / 111 - 130 km/h / 30 - 36 m/s
Climb Rate	365 meters per minute / 1,200 feet per minute
Turn Radius	Approx. 180 m / 590 ft
RANGE	
Maximum Ferry Distance	9.5 km / 6.0 miles
Endurance	30 minutes (+10 minute reserve)
RUNWAY	
Take-off run with Max Payload	150 m / 500 ft
Required Runway	200 m x 10 m / 656 ft x 33 ft
Surface	Paved, gravel, dirt, grass
Maximum demonstrated crosswind	18 knots / 8 m/s
SPRAY PERFORMANCE (Inclusive of ferry, fill, battery swaps)	
Work Rate Maximum (@ 18 m / 59 ft swath width)	81 ha/hour (200 acres/hour) @ 24 l/ha 98 ha/hour (242 acres/hour) @ 10 l/ha
FEATURES	
Fully Autonomous	Minimal training required to operate
Safety	- Fully redundant propulsion, controls, and sensor suite - Backup parachute system - Dump Door
Night flying	GPS and Laser/RADAR based navigation



Pelican Spray prepares for night flight mission.

3. DRAWINGS



4. SPRAY SYSTEM & PERFORMANCE



PERFORMANCE

Wind compensation

Low application speed

Variable rate application

Real-time gust inversion & detection

450 micron shear-point



SPRAY SYSTEM

Spray Nozzles

24 Hollow-cone Hydraulic

Continuously variable droplet size adjustment

Variable Speed Rotary Atomizers

Narrower droplet spectrum distribution

5. GROUND CONTROL STATION



6. OPERATION & TRAINING

OPERATION

Fully autonomous operation

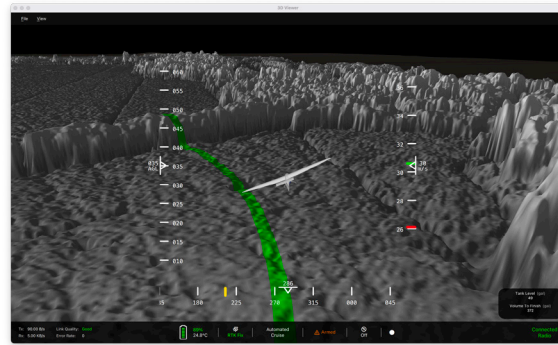
Flight plan programming by single on-ground pilot

Dynamic rerouting via satellite link

Fill cap avoiding chemical spillage

Chemical tank spill (< 60 sec)

On-ground battery swap (< 5 min)



TRAINING

Four week pilot training

- Two week simulator training
- Two week aircraft flight training

7. CHARGING INFRASTRUCTURE

GROUND CHARGING STATION	
Grid-charging three phase	220 V, 30 Amps
Generator	Diesel

8. GROUND HANDLING

ITEM	DESCRIPTION
Trailer Transportation	Standard size, custom holding brackets Requires simple wing removal
Hangar Storage	40 ft HC sea container
Disassembly / Assembly	Under 15 mins
Daily Chemical Tank Rinse	Under 20 mins



9. MAINTENANCE

ITEM	INTERVAL / COST
Battery	Replacement est. every 2,000 flight hours [Contact Pyka Sales for details]
Motors / Propellers	On condition [Contact Pyka Sales for details]
Landing Gear	On condition
Total Maintenance Cost	Est. around \$14.50 / Flight Hour *Assuming 200-hour inspection interval, battery and general parts replacement

10. FINANCIAL

CAPITAL COST	
Annual Lease Rate	Replacement est. every 2,000 Flight Hours [Contact Pyka Sales for details]
Warranties	Standard 12-18 months warranties on all parts Optionality to include all parts replacement and insurance during the lease term
List Price	[Contact Pyka Sales for details]
Economic Useful Life	Est. four (4) years
Financing Options	[Contact Pyka Sales for details]

11. DEVELOPMENT TIMELINE

FLIGHT	
First Flight	March 2020
Proving / demo flights	Q4 2021 / Q1 2022
Entry into Service	Q1 2022
CERTIFICATION	
United States	FAA Special Airworthiness Certificate • Granted October 2020 - Renewed in December 2022 - Renewed in August 2023 • R&D / Demo flights
	FAA 44807 Exemption • Approved in July 2023
New Zealand	Based on 'Egret' certification granted in 2019 • Classified as Class 3
Costa Rica	Received full DGAC approval in September 2022, including: • Commercial Chemical Application • Night Flight Operation
Honduras	Commercial trial operations started in early Q1 2023
Guatemala	Application raised for commercial flight approvals in Q2 2022
Brazil	Special Airworthiness Certificate for chemical spray and EVLOS operation granted in January 2022 Certification granted in April 2023 as part of Class 3 RPAS regulation.

12. IMAGERY





13. CONTACT

For further information, please contact the manufacturer:

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